

FULL NAME: _____

BLOCK: _____

Inverse Variation — Introduction

Lesson 4-1 · Period 1 · Teacher Edition

OBJECTIVES

Math Objective	Identify when two variables vary inversely by testing whether their product xy is constant, then write the inverse-variation equation $y = k/x$ and use it to predict unknown values.
Language Objective	Students will use the frame “As ___ increases, ___ decreases; their product is ___, so the equation is $y = ______ / x$.” to explain an inverse-variation relationship.
Essential Understanding	Two variables vary inversely when their product is a constant k . The equation $y = k/x$ is the algebraic form; $xy = k$ is the structural form. Graphically, the relationship is a hyperbola (the reciprocal function).

TOPIC GOALS / ESSENTIAL QUESTION / MATERIALS

Topic Goals	Students will distinguish inverse variation from direct variation using product vs ratio tests, and write the equation $y = k/x$ from two given values. All student-facing items are Savvas-sourced (bank IDs in teacher edition).
Essential Question	How are inverse variations related to the reciprocal function?
Materials	Pencils; student packet; calculator (optional). No Chromebooks required for Period 1.

[PIN] PRE-FLIGHT NOTE (read before class)

Explore is 7 bank items: Try-It 1 → #9 → #14 → Try-It 2 → #13 → #16 → #10 (extension). Paced for ~4–5 min per item.

If pairs finish early, push them to the Optional Reinforcement: Model & Discuss Parts B + C (verbal/TPS), then Practice #15.

If pace slows, cut #10 (Construct Arguments) — it’s the DOK-2 stretch; the core DOK-1 arc (Try-It 1 → #9/#14 → Try-It 2 → #13/#16) is the minimum viable set.

Tag: notice & wonder

Do Now

DOK: 2

Minutes: 5

Teacher does

- Project the two rectangles (or use printed Do Now sheet).
- Circulate silently. Do NOT teach during this phase.
- Collect at 5 min sharp.

Students do

- Write 1 notice + 1 wonder.
- Sketch a third rectangle with area 144.

Questions to ask

- (None during the phase — teacher harvests answers in Launch.)

Adult role

Solo teacher: circulate silently. Resist the urge to give hints.

[RECT] TWO RECTANGLES — Both have area = 144 square units

Rectangle 1: length = 72, width = 2

Rectangle 2: length = 24, width = 6

A) Notice: Write something you notice about these two rectangles.

B) Wonder: Write something you wonder about rectangles with area 144.

C) Sketch: Draw a third rectangle with area 144. Label its length and width.

Do Now (4-1-savvas-model-discuss-lesson-4-1-launch)

The two rectangles shown both have an area of 144 square units (72×2 and 24×6). (A) Sketch as many other rectangles as you can that have the same area. Organize and record your data for the lengths and widths of the rectangles. (B) Use Structure: Considering rectangles with an area of 144 square units, what happens to the width of the rectangle as the length increases? (C) Examine at least five other pairs of rectangles, each pair sharing the same area. How would you describe the relationship between the lengths and widths?

[CHECK] ANTICIPATED STUDENT RESPONSES

NOTICE (typical): “Both rectangles have the same area.” “One is long and thin, the other is shorter and wider.” “When the length is bigger, the width is smaller.”

WONDER (typical): “Can any rectangle have area 144?” “What’s the rule?” “Is there a formula?”

SKETCHES: students might draw 1×144 , 4×36 , 8×18 , 12×12 , 3×48 , etc. All valid.

KEY MOVE in Launch: surface the insight that $\text{length} \times \text{width} = 144$ for every rectangle — this is $xy = k$.

LAUNCH — Inverse Variation (teacher models on screen)

[BOOK] INVERSE VARIATION RULES (keep printed on your page)

- Two variables vary INVERSELY when their product is a constant k :
 $xy = k$ or equivalently $y = k/x$
- The constant k is the CONSTANT OF VARIATION.
- As one variable INCREASES, the other DECREASES proportionally.
- Example: if $x = 10$ and $y = 3$ vary inversely, then $k = xy = 30$.
Equation: $y = 30/x$. When $x = -6$: $y = 30/(-6) = -5$.

[WARN] COMMON ERROR

Do NOT write $y/x = k$ — that is DIRECT variation.
Inverse variation multiplies, direct variation divides.
Test: compute xy for several points. If constant $\rightarrow$ inverse.

Tag: teacher models Example 1 + Example 2

Launch

DOK: 1–2

Minutes: 12

Teacher does

- Bridge from Do Now: “You all noticed $\text{length} \times \text{width} = 144$. That’s a RULE, and it has a name: inverse variation.”
- Project Example 1 (Savvas Topic 4-1). Think aloud: for each table, compute xy for each column. Is it constant? $\rightarrow$ YES = inverse; NO = not.
- Project Example 2. Think aloud: given $x = 10$, $y = 3$, find $k = xy = 30$. Write $y = 30/x$. Substitute $x = -6$. Answer: $y = -5$.
- Pause $1\times$ for 30-sec partner-check: “What’s k for your Do Now rectangles?” (Answer: $k = 144$.)
- Do NOT solve Try-Its. Release promptly at minute 17.

Students do

- Watch silently through Example 1.
- During partner-check: “What’s k for area = 144?” (144)
- Copy the Inverse Variation Rules card into their page margin.
- Note the COMMON ERROR warning: $y/x = k$ is DIRECT variation.

Questions to ask

- “What’s xy for column 1? Column 2? Column 3? Is it constant?”
- “If two variables have $xy = 30$ always, what is k ?”
- “How do I write y in terms of x if $xy = 30$?”

Adult role

Project, think aloud. Do NOT give students Try-Its to work until minute 17.

[MIC] TEACHER SCRIPT

1. “From the Do Now, you noticed $\text{length} \times \text{width} = 144$ for every rectangle. That’s inverse variation — a pattern.”
2. Example 1 walk-through: “Watch me. For each column, I compute xy . If it’s the same number for every column, the two variables vary inversely.”
3. Example 2 walk-through: “Now I’ll write the EQUATION. $k = xy$, so if $x = 10$ and $y = 3$, then $k = 30$. The equation is $y = k/x$, so $y = 30/x$. When $x = -6$, $y = 30/(-6) = -5$.”
4. Common-error flag: “Don’t mix up $y/x = k$ (direct) with $xy = k$ (inverse). Inverse multiplies.”
5. Release: “Now you try. Try-It 1 asks if two tables are inverse variation. Try-It 2 asks you to write the equation.”

EXPLORE — Think-Pair-Share (33 min)

Work with a partner. Use the Inverse Variation Rules above for every problem. Move through items in order — each builds on the last.

Tag: Think-Pair-Share + Practice

Explore

DOK: 1–2

Minutes: 33

Teacher does

- 5 laps circulation across 33 min (~6–7 min each).
- Lap 1: Try-It 1 — press pairs to compute xy for every column.
- Lap 2: #9 + #14 — check that pairs notice NOT every table is inverse.
- Lap 3: Try-It 2 — press pairs to state k , then write the equation.
- Lap 4: #13 + #16 — quick check on generalization and application.
- Lap 5: #10 (Construct Arguments) — stretch. Cut if pace slows.
- Do not give answers. Use sentence frame to press justification.

Students do

- Pairs move in order through 7 items: 2 Try-Its + 5 Practice.
- For each table (#14, #15 later): compute xy for EVERY column before answering.
- For Practice #10 (why can $x = 0$ not be in the domain?): justify with the equation $y = k/x$.
- Justify with sentence frame before writing.

Questions to ask

- What's xy for each column? Is it constant?
- If xy is NOT constant, what kind of variation is it? (Check: is y/x constant = direct variation.)
- What's the value of k ? How do you know?
- Can you write the equation $y = k/x$?
- For #10: if $y = k/x$ and $x = 0$, what happens? (Division by zero = undefined.)

Adult role

Solo teacher: 5 laps. Press pairs to justify using the rule card. No answers.

Bank-item labels (in order):

1. Try It 1 — Identify inverse variation
2. Practice #9 — Generalize
3. Practice #14 — Another table check
4. Try It 2 — Write the equation
5. Practice #13 — Generalize: write k in terms of x and y
6. Practice #16 — Apply the equation

7. Practice #10 — Construct Arguments (extension)

Try It 1 — Identify inverse variation

Try It 1. Determine if each table of values represents an inverse variation.

(a) $x : 1, 2, 3, 5, 6, 15$; $y : 25.5, 12.75, 8.50, 5.10, 4.25, 1.70$.

x	1	2	3	5	6	15
y	25.5	12.75	8.50	5.10	4.25	1.70

(b) $x : 6.6, 5.5, 4.4, 3.3, 2.2, 1.1$; $y : 3, 5, 7, 9, 11, 13$.

x	6.6	5.5	4.4	3.3	2.2	1.1
y	3	5	7	9	11	13

[CHECK] ANSWER / ANTICIPATED TALK

Answer key: (a) all products = 25.5 $\rightarrow$ yes, inverse variation; (b) products not constant ($6.6 \cdot 3 = 19.8$; $5.5 \cdot 5 = 27.5$; ...) $\rightarrow$ no. Use during Explore as check-for-understanding after Example 1.

Practice #9 — Generalize

9. Generalize: Just from looking at the table of values, how can you determine that the data do NOT represent an inverse variation? Table - $x : -2, 2, 4, 6, 8, 10$; $y : -6, 6, 12, 18, 24, 30$.

x	-2	2	4	6	8	10
y	-6	6	12	18	24	30

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from TE): As the x -values increase, the y -values also increase. (Additionally, this IS a direct variation with $k = 3$ since $y/x = 3$ for every pair.)

Practice #14 — Another table check

14. Do the tables of values represent inverse variations? Explain. See Example 1. Table - $x : -1/4, -1/2, 1/3, 2, 5, 11$; $y : -9/2, -9, 6, 36, 90, 198$.

x	$-\frac{1}{4}$	$-\frac{1}{2}$	$\frac{1}{3}$	2	5	11
y	$-\frac{9}{2}$	-9	6	36	90	198

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from TE): No; as the x -values increase, the y -values also increase. (This is a direct variation, where $k = 18$ since $y/x = 18$ for every pair.)

Try It 2 — Write the equation

Try It 2. In an inverse variation, $x = 6$ when $y = 1/2$. (a) What is the equation that represents the inverse variation? (b) What is the value of y when $x = 15$?

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (confirmed by teacher from TE): (a) $y = 3/x$ (since $k = 6 \cdot (1/2) = 3$); (b) when $x = 15$, $y = 3/15 = 1/5$.

Practice #13 — Generalize: write k in terms of x and y

13. Generalize: For an inverse variation, write an equation that gives the value of k in terms of x and y .

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from TE): $k = xy$.

Practice #16 — Apply the equation

16. If x and y vary inversely and $x = 3$ when $y = 2/3$, what is the value of y when $x = -1$?

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from TE): When $x = -1$, $y = -2$. IMPORTANT: the original screenshot had the y -value in the setup cut off at the line break. Reconstructed as $y = 2/3$ from answer key ($k = 3 \cdot (2/3) = 2$; at $x = -1$, $y = -2$ checks). Visual_needs_cleanup=true flags that the source screenshot is incomplete.

Practice #10 — Construct Arguments (extension)

10. Construct Arguments: Explain why zero cannot be in the domain of an inverse variation.

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from TE): Division by zero is undefined. (Elaboration: since $y = k/x$, the value $x = 0$ produces $y = k/0$ which is undefined, so $x = 0$ must be excluded from the domain.)

SHARE / SUMMARY (5 min)**Sentence frames**

Pair share: “As ___ increases, ___ decreases. Their product is ___, so the equation is $y = ___ / x$.”

Whole class: one pair reads their sentence. Class signals thumbs up/sideways.

Tag: *academic conversation*

Share / Summary

DOK: 2

Minutes: 5

Teacher does

- Cold-call 2 pairs to read their sentence-frame sentence.
- Write one student-generated sentence on board.
- Preview P2: “Next class we’ll solve real-world inverse-variation problems — like a bouzouki string.”

Students do

- Presenting pair reads their sentence.
- Class signals thumbs up/sideways.
- Note: next class applies this to a real situation.

Questions to ask

- What's a sentence that captures today's rule?
- What's the difference between inverse and direct variation?

Adult role

Cold-call 2 pairs. Keep it brief — 5 min.

Tag: summary recap (not graded)

Exit Ticket

DOK: 1-2

Minutes: 2

Teacher does

- Collect at door.
- Scan #2 (inverse vs direct) — misconceptions tell you what to re-hit in P2 Do Now.

Students do

- Complete 3 sentence stems.
- Be honest about confusion.

Questions to ask

- Did students correctly distinguish inverse ($xy = k$) from direct ($y/x = k$)?

Adult role

Collect. No grade.

[CLIPBOARD] EXIT TICKET — Student Form

Today I learned that ____.

The difference between inverse and direct variation is ____.

One thing I'm still unsure about is ____.

OPTIONAL REINFORCEMENT (not collected)

If you finish Explore early or want extra practice before Period 2.

[SEARCH] OPTIONAL • Model & Discuss Parts B + C

Return to the two rectangles from the Do Now (both have area 144). With your partner:

B) Use Structure. Considering rectangles with area 144, what happens to the WIDTH as the LENGTH increases?

C) Examine at least five more pairs of rectangles with area 144. How would you describe the relationship between their lengths and widths — in your own words?

Optional #1 (Savvas Practice)

15. Do the tables of values represent inverse variations? Explain. See Example 1. Table - x : 1, 2, 3, 4, 5, 6; y : 60, 30, 20, 15, 12, 10.

x	1	2	3	4	5	6
y	60	30	20	15	12	10

[CHECK] ANSWER / ANTICIPATED TALK

Answer key (from TE): Yes; as the x -values increase, the y -values decrease. Also the product of each pair of x - and y -values is 60 ($k = 60$, so $y = 60/x$).

[CLOCK] PERIOD A & F — 65-MIN CLASS NOTES

Both sections get 65 min for P1 (Fri F / Mon A). Use the extra 10 min on Explore, NOT Launch. Suggested add: project Savvas Practice #9 and #14 on screen (not in packet) for 2 more TPS rounds. Or: extend Model & Discuss Parts B + C with “find the equation in the form $y = k/x$ for your three rectangles.” ($k = 144$, so $y = 144/x$.)

[PUZZLE] MODIFICATIONS / SUPPORT FOR IEP STUDENTS

- Provide the Inverse Variation Rules card as a sticker/cut-out on their desk.
- For Try-It 1, allow students to compute xy with a calculator for every column.
- Accept verbal sentence-frame completions in lieu of written.

[GLOBE] SUPPORTS FOR ELL STUDENTS

- Sentence frames printed on student packet (don't skip).
- Word card: “varies inversely” = as one goes up, the other goes down proportionally.
- “constant of variation” = k = the fixed product xy .
- “product” = the result of multiplying two numbers.
- Pair ELL students with English-dominant partners for TPS.